Malthus’s bleak vision did not happen, thanks to technological advances in agriculture and the development of artificial fertilizers, but his population model became applicable to species populations within ecosystems. Every habitat, and the niche occupied by a species within its community of organisms, has a carrying capacity—the maximum population that can be supported by the resources available, such as water, space, food, and light. Any rise in population above this level is likely to be reduced by naturally occurring factors. As a result, wild populations should in theory be more or less static, fluctuating only around the carrying capacity, assuming the random impacts of catastrophic events are ignored.

However, this relative equilibrium did not always match up with observations—as in D’Ancona’s account of a sudden increase in the population of predatory sea fish. One theory to explain this discrepancy started from the premise that the population of predators is related to the size of the population of their food supply, such as prey species. The relationship suggests that when a lot of food is available, there will be a large predator population. The growing predator population should then begin to reduce the amount of prey, which will in turn lead to a drop in the number of predators. The size of both populations will rise and fall, but the ratio of predators to prey will remain stable.

Such a balanced theory was still at odds with species observations. Through mathematical modeling, Volterra was able to show that the average sizes of predator and prey populations do indeed oscillate but the rate at which each population is growing or declining is always changing and almost never matches the changes experienced by the other population. To eliminate variables, Volterra made a series of assumptions: first, that the prey and predator species have no reproduction limits and the rate of change in a population is proportional to its size; second, that the prey population—presumed to be a herbivore—is always able to find enough food to survive. Next, they assumed that the prey population is the